

GPU Server

This page is meant to summarize information about our own GPU server.

See [GPU-Server Management repo](#) and its Wiki for information and Ansible scripts for server administration and management.

User Guide

Whenever you operate on the server, please make sure to block it by adding an entry to [this spreadsheet](#).

Connecting to server

1. Connect to university network directly or via VPN
2. In a shell type: `ssh <user>@pc12830.mathematik.uni-marburg.de` and enter your password
3. You can add your SSH key to the server to skip password entry (see next section)

Add SSH key

1. [Generate an SSH key on your local machine](#).
2. Copy your public key to the server as authorized key: `scp ~/.ssh/<key>.pub <user>@pc12830.mathematik.uni-marburg.de:~/.ssh/authorized_keys` (note: if you want multiple authorized keys, you need to add all of them to `~/.ssh/authorized_keys` on the server instead of copying a single key and renaming it)

Working with Repositories

1. [Create an SSH key for your user on the server](#).
2. [Add it as deploy key to your repository](#).
3. You should now be able to checkout and pull/push commits to the repository where you added the deploy key.

User Commands

There are some sudo commands which can be run as no-sudo user. Run the `syscmd` command to get a selectable list of these commands.

Clean experiment environment

To obtain a clean experiment environment do the following:

1. Block the server by entering your name in [this google spreadsheet](#)
2. Stop the LLM service ([see LLM below](#))

Please remember to restart all services you took offline to get your experiment environment after your experiments have finished.

Summary

parameter	value
IP address	137.248.184.1
hostname	pc12830.mathematik.uni-marburg.de
OS	Ubuntu 22.04 LTS
SSH port	22
credentials	FB12 Linux

Storage

Available storage

kind	size	purpose	RAID	mountpoint
SSD	225 GB	OS	1	?

kind	size	purpose	RAID	mountpoint
SSD	1.3 TB	storage	0	/data
SATA	9.1 TB	storage	1	/storage

Location of /home

The server is tied to the faculty's LDAP. For this reason the "normal" home folders of the faculty's servers can not be mounted to the GPU server.

At a user's first login on the GPU server a local home directory will be created at `/app/home/` , which happens to be the LDAP home directory. The directory `/app/home/` is linked to `/storage/home/` . So the homes actually reside on `/storage` .

Network

Our GPU server can be reached via SSH over the IP-address `137.248.184.1` on port 22. Its hostname is `pc12830.mathematik.uni-marburg.de` . To access the server the FB12 Linux credentials are required.

The `ufw` firewall should be active but allow connections via SSH and HTTPS from within the university network. SSH connections from outside the university's network are being refused.

To establish an SSH connection from outside the university network a VPN connection is required. Alternatively, `login.mathematik.uni-marburg.de` can be used as jumphost.

Backups

The partitions `/data` and `/storage` are being incrementally backed up on weekday nights. Deleted files or previous versions of files are being stored for 90 days by the HRZ on their backup server.

The backup client can be started/stopped via `systemctl (start|stop) dsmcad` . Its configuration files `dsm.opt` and `dsm.sys` can be found in `/opt/tivoli/tsm/client/ba/bin` . Email from HRZ regarding backups is being sent to the faculty's `itadmin@...` address and forwarded to Alex (vielhau4).

Services

LLM

The GPU server hosts an LLM service managed by [this repository](#). You can start and stop the service via: `sudo service llm {start,stop}` and check its status via `sudo service llm status` . Note that you can run these commands even as no-sudo user and they are selectable from the list of no-root user commands when you run `syscmd` .

FURTHRmind

[See here for details](#).